

DevOps + AI/ML = MLOps Roadmap (Compact)

STAGE 0: Prerequisites

- Python, Git, Linux basics

STAGE 1: Core DevOps Skills

- Git & GitHub: clone, commit, push, branch
- CI/CD: GitHub Actions, GitLab CI, Jenkins
- Env Mgmt: virtualenv, conda, Docker
- IaC: Terraform, AWS CloudFormation

STAGE 2: ML-Specific DevOps (MLOps)

- Experiment Tracking: MLflow, DVC, W&B
- Model Packaging: joblib, pickle, tf.saved_model
- Model Serving: Flask, FastAPI, TF Serving, TorchServe
- Cloud & Orchestration: AWS Sagemaker, GCP AI, Docker, K8s, Kubeflow

STAGE 3: Advanced MLOps

- Monitoring: Prometheus, Grafana, ELK, WhyLabs, Evidently
- Automation: Airflow, DVC, Kubeflow Pipelines
- Retraining Pipelines and Model Rollbacks

FULL SUMMARY

1. Basics: Git, Python, Linux
2. DevOps: CI/CD, Docker, Terraform
3. ML Integration: Packaging, APIs, MLflow
4. Production: K8s, Serving, Cloud
5. Pipelines: Automation, Monitoring

LEARNING RESOURCES

- Git: learngitbranching.js.org
- Docker: Docker docs, Docker for Data Science
- CI/CD: GitHub Actions Docs, Real Python
- MLflow: mlflow.org | DVC: dvc.org
- Kubernetes: kubernetes.io | FastAPI: fastapi.tiangolo.com
- Kubeflow: kubeflow.org